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Systematic Reviews in the Sciences

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Frameworks

A framework provides a structure which can help you to clearly define your research question in terms of population, intervention or phenomenon of interest, comparisons, settings, outcomes and more.

Common frameworks include PICO, SPIDER and SPICE.

PICO

The PICO framework is most suitable for clinical and intervention based research questions. If you plan to review prognostic or qualitative data, or diagnostic test accuracy, another

framework, such as SPIDER or SPICE, may be more suitable.

Common variations on PICO can include:

- PICOT = Population, Intervention, Comparison, Outcome, Type of Study or Time within which the outcome will be observed
- PPICO = If your population is more complex, it may need two descriptors for its population.
- PI_O = Your question may not need a comparison.
- PCC = Population/Problem, Concept, Context (often used for Scoping Reviews)
- PEO = Population, Exposure, Outcome.

PICO Framework		
P	Patient, Population or Problem	What are the most important characteristics of the patients? (e.g. gender, age, and disease or condition)
I	Intervention or Exposure	What is the main intervention? (e.g. drug treatment, diagnostic and screening test)
C	Comparison or Control	What is the main alternative to the intervention? (e.g. placebo, different drug, surgery, no treatment)
O	Outcome	What are you trying to accomplish improve, measure, effect? (e.g. quality of life, morbidity, complications)

SPIDER

The SPIDER framework is useful for qualitative/mixed methods research.

SPIDER Framework		
S	Sample	What is the group of people you are studying? (e.g. young parents, first year university students)
PI	Phenomenon of Interest	What is the topic of your research?
D	Design	How was the study conducted? (e.g. focus group, interview, observational study, survey)
E	Evaluation	What is the measurement of outcome of the study? This might be subjective and not necessarily empirical (e.g. experiences)
R	Research Type	Was it qualitative or mixed research?

SPICE

The SPICE framework is useful for qualitative research topics evaluating the outcomes of a service, project or intervention.

SPICE Framework		
S	Setting	What is the location or environment of the research or study? (e.g. South Korean Universities, a hospital emergency room)
P	Perspective	Who is the group being studied? (e.g. adolescents, rural doctors)
I	Intervention	What is the intervention/interest/exposure that you are evaluating?
C	Comparison	What is the alternative strategy you would like to compare to the intervention? May include no intervention
E	Evaluation	What is the result or outcome measures? (e.g. reduced anxiety, improved memory)